IS 108 – INTELLIGENCE SYSTEM FINAL PROJECT SY 2025-2026

BUSINESS INTELLIGENCE PREDICTIVE MODELING APPLICATION

I. Final Project Description

Work in a team with a maximum of 3 members to design and develop a Business Intelligence application that performs a complete predictive modeling process using real or realistic business-related data.

The application must allow users to input or load data, process the dataset, train predictive models, compare results, and display predictions in a user-friendly interface.

II. Project Goal

Create an application that can help a business or organization make better decisions by predicting outcomes using machine learning.

III. Required Algorithms

The application must implement and compare the following three algorithms:

1. K-Nearest Neighbor (KNN)
2. Support Vector Machine (SVM)
3. Artificial Neural Network (ANN)

IV. Suggested Business Problem Areas

Each team may choose one business-related prediction problem such as:

1. Customer churn prediction
2. Product demand prediction
3. Loan approval prediction
4. Employee attrition prediction
5. Sales performance classification
6. Fraud detection
7. Student performance prediction
8. Marketing campaign response prediction

V. Required Features of the Application

The application must include the following:

1. Dataset Handling

- ✓ Import dataset from CSV or Excel
- ✓ Display dataset in tabular form
- ✓ Show basic information about the dataset

2. Data Preprocessing

- ✓ Handle missing values
- ✓ Encode categorical data if needed
- ✓ Perform feature scaling when necessary
- ✓ Split dataset into training and testing sets

3. Predictive Modeling Process

The application must demonstrate the predictive modeling workflow:

- ✓ Problem identification
- ✓ Data collection

- ✓ Data preprocessing
- ✓ Feature selection
- ✓ Model training
- ✓ Model testing
- ✓ Performance evaluation
- ✓ Prediction output

4. Model Implementation

The application must train and test all three models:

- KNN
- SVM
- ANN

5. Model Evaluation

The application must show model performance using appropriate metrics such as:

- ✓ Accuracy
- ✓ Precision
- ✓ Recall
- ✓ F1-score
- ✓ Confusion matrix

VI Required Deliverables

Each group must submit the following:

1. Working Application

A functional predictive modeling application.

2. Source Code

Complete and organized source code with comments.

3. Project Documentation

A short project paper or documentation containing:

- Title of the project
- Members of the group
- Business problem addressed
- Dataset used
- Data preprocessing steps
- Description of KNN, SVM, and ANN implementation
- Evaluation metrics used
- Comparison of results
- Conclusion and recommendations

4. Presentation

A 10–15 minute presentation with live demo of the application.

VII. Grading Rubric

Total: 100 Pts

I. Application Functionality – 25 points
II. Correct Implementation of KNN, SVM, and ANN – 20 points
III. Predictive Modeling Process – 15 points
IV. User Interface Design and Usability – 10 points
V. Model Evaluation and Comparison – 15 points
VI. Documentation/Project Report – 10 points
VII. Presentation and Demonstration – 5 points